Parallel Coordinate Analysis of the Four Fundamental Interactions

Working Draft

May 19, 2026

Abstract

This draft consolidates a parallel-coordinate visualization of the four fundamental interactions with a short commentary on the static strength values, presentation caveats, and the running of the Standard Model gauge couplings. The static snapshot is contrasted with the renormalization group evolution of the couplings, which is where the unification picture actually lives.

1 Parallel Parameter Analysis

The two figures (parallel coordinate matrix and linear parallel scales) summarize the four interactions across five descriptive variables.

Variable 1 (Interaction Identity). Defines the four pillars of modern quantum and relativistic particle dynamics: the strong, electromagnetic, weak, and gravitational interactions.

Variable 2 (Gauge Boson Carrier). The intermediate vector particles responsible for transmitting interaction momentum between fields: the gluon, photon, W and Z bosons, and the hypothetical graviton.

Variable 3 (Relative Strength Constant). Shows the massive separation of force scales, spanning many orders of magnitude between mass geometry (gravity) and color-charge confinement (strong force).

Variables 4 and 5 (Range Profiles). Contrasts the localized forces bounded by physical thresholds ($\lesssim 10^{-15}$ to 10^{-18} m) against fields with zero-mass gauge bosons that maintain an infinite geometric reach.

Interaction	Gauge Boson	Relative Strength	Range Type	Boundary Scale
Strong	Gluon	1 (reference)	Short range	$\lesssim 10^{-15}$ m
Electromagnetic	Photon	10^{-2}	Infinite reach	Infinite
Weak	W, Z bosons	10^{-13}	Short range	$\lesssim 10^{-18}$ m
Gravitational	Graviton*	10^{-38}	Infinite reach	Infinite

Table 1: Standard summary of the four interactions. The asterisk on the graviton marks it as a hypothetical mediator that has not been observed.

2 Caveats on the Static Picture

The summary above is accurate as a textbook snapshot but hides three things worth flagging.

Energy dependence of the weak coupling. The often-cited ratio of 10^{-13} for the weak force is the low-energy figure at roughly proton-scale distances. At electroweak unification energies (around 100 GeV) the weak coupling approaches the electromagnetic one, which is the entire point of electroweak theory. A static column in a comparison plot necessarily hides that energy dependence.

Consistent labeling of the graviton. The graviton carries an asterisk in the parallel coordinate matrix but not in the linear parallel scales plot. It is the only entry that has never been observed, so the marker should be present in both figures or appear in a shared footnote.

Strength normalization convention. Setting the strong force to 1.0 as the reference is a convention. Some sources normalize against the electromagnetic coupling instead, which shifts every other entry by two orders of magnitude. The reference choice should appear in the caption so the reader is not left to guess.

Log scaling honesty. The linear parallel scales figure explicitly labels its axes as $\log_{10}$ of the relevant quantity, which faithfully renders the 38 orders of magnitude separating gravity from the strong force. The parallel coordinate matrix shows the raw values like 10^{-38} at uniform vertical spacing, which is not visually proportional and can be misleading at a glance. The lower figure is the honest one.

3 Running Couplings and Where Unification Lives

The static strength values are snapshots at a specific reference energy, usually near the proton mass scale. The actual couplings depend on the momentum transfer Q of the interaction and evolve according to the renormalization group equations.

For the three Standard Model gauge couplings, the one-loop running behaves as follows.

- The strong coupling α_s decreases with energy. This is asymptotic freedom. At the Z mass ($M_Z \approx 91\,\text{GeV}$) it sits near $\alpha_s(M_Z) \approx 0.118$. By $Q \sim 10^{15}\,\text{GeV}$ it has dropped to roughly 0.02.
- The electromagnetic coupling α_{em} runs the other way. At low energies it is the familiar 1/137. At the Z mass it has risen to roughly 1/128.
- The weak SU(2) coupling α_w runs slowly downward with increasing energy.

When $1/\alpha_i$ is plotted against $\log Q$, the three lines move toward each other. In the minimal Standard Model they almost meet near $10^{15}\,\text{GeV}$ but miss. In supersymmetric extensions they meet much more cleanly near $2 \times 10^{16}\,\text{GeV}$, which is one of the historical motivations for supersymmetry. This is the grand unification picture.

Gravity does not fit on the same plot in any controlled way. The naive estimate is that the gravitational coupling becomes order one near the Planck scale, $M_{\text{Pl}} \sim 10^{19}\,\text{GeV}$, which is why people sometimes draw it crashing into the others up there. Without a working quantum theory of gravity, that line is hand-waving.

4 Next Steps

A natural companion figure to the two static diagrams is a plot of $\alpha_i^{-1}(Q)$ for $i = 1, 2, 3$ running from M_Z up to roughly $10^{18}\,\text{GeV}$, with both Standard Model and MSSM beta function coefficients overlaid. This shows the unification difference between the two scenarios side by side and provides a more honest dynamic complement to the static comparison.